

Project Statement

Project Title

Hospital Patient Management System

Problem Statement

Hospitals need a quick and reliable way to record and retrieve basic patient details without relying on paper registers or scattered files. Manually maintained records make it difficult to add new patients, look up existing ones, or remove outdated entries in an organized way.

The Hospital Patient Management System is a simple command-line Python program that combines patient record creation, viewing, searching, and deletion with permanent storage in a CSV file.

Scope

The project is designed for basic hospital record-keeping. It allows the user to:

- Add a new patient record.
- Display all stored patient records.
- Search for a patient using their Patient ID.
- Delete a patient record.
- Store patient data permanently in a CSV file (patients.csv).

The project does not use a GUI, a database engine, an internet connection, or external Python packages beyond the standard library.

Target Users

- Small clinics or hospital front-desk staff needing a lightweight record tool.
- College and school students studying Python file handling.
- Beginners learning Python.

Major Functional Modules

1. Patient Record Management – add, display, search, and delete patient records.
2. Data Storage Handling – reads from and writes to the patients.csv file, creating it automatically if it does not exist.

hospital_management.py contains the command-line entry point, the menu controller, and the functional logic together, since the project is implemented as a single script rather than split into separate module files.

Menu Options

- Add Patient
- Show Patients
- Search Patient
- Delete Patient
- Exit

Data Fields Stored

- Patient ID
- Name
- Age
- Gender
- Disease
- Doctor
- Room

Technology

- Python 3
- Python standard library (csv module, os module)
- CSV text file for local storage
- Git and GitHub for version control

Objective

The main objective of this project is to demonstrate how basic Python programming and file handling can be used to build a simple, real-world hospital record management application.

Future Improvements

The project can be extended by adding:

- Patient update functionality
- Appointment management
- Billing system
- Doctor management
- Login authentication
- Graphical user interface
- Database connectivity using SQLite or MySQL

Disclaimer

This is an educational first-year engineering project and is not intended for use as a production hospital information system.